A Benchmark Study of Formula-Aware Document Read

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Abstract

We present a benchmark for formula-aware document reading. The proposed method combines native text extraction with vision-language recognition, achieving high fidelity on equations and tables while preserving paragraph structure for machine translation.

1. Introduction

Recent advances in vision-language models have improved document parsing. However, most systems still struggle with reading order in multi-column layouts. In this work, we study how to preserve the semantic structure of academic papers when converting PDF to HTML. Our key insight is that translation quality depends on paragraph-level DOM structure rather than pixel fidelity. We evaluate the approach on a corpus of 20 papers across computer science and physics.

The rest of this paper is organized as follows. Section 2 describes the pipeline. Section 3 reports experimental results on formulas and tables. Section 4 discusses limitations and future work.

2. Method

Our pipeline first performs preflight checks to detect whether a page contains native text. Pages with insufficient text are routed to a full OCR pipeline. For native pages, we extract embedded text directly and analyze the layout to locate formulas and tables. Formula regions are cropped and recognized into LaTeX, which is rendered with MathJax in the final HTML.

3. Results

We found that native text extraction achieves a character accuracy of 99.2 percent. Formula recognition reached 96 percent accuracy on the test set. The reading order was correct for 97 percent of multi-column pages.

4. Equations

The energy-momentum relation is given by the following equation, which is central to relativistic mechanics and appears in many physics papers.

$$E^2 = (pc)^2 + (mc^2)^2$$

5. Tables

Table 1 summarizes the benchmark results across the three methods.

Method	Text Acc	Formula Acc	Order
Ours	99.2%	96.0%	97%
Baseline A	97.5%	91.0%	90%
Baseline B	96.1%	88.5%	86%

6. References

[1] Smith J, et al. Formula-aware document parsing. In Proc. ACL 2025.

[2] Doe A. Reading order recovery for two-column layouts. arXiv:2601.00001.

[3] Chen X. MathJax rendering in translated documents. 2024.

7. Figures

Figure 1 shows the overall pipeline of the proposed system.



